CALIBRATION SHEETS

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SBE 37IM Temperature Calibration - S/N 3883..... 3

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CALIBRATION SHEETS

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SBE37-IM MicroCAT

Conductivity & Temperature Recorder
with Inductive Modem

Instrument Configuration:

Serial Number	37IM39208-3883
Pressure Sensor	2000 dBar Druck, SN 6998
Firmware Version	2.3a
Memory	2048Kb
Conductivity Range	0-7 S/m
Microcat ID#	01
DPSK Baud Rate	1200 (Microcat <-> SIM)
SIM Baud Rate	9600, 8 data bits, 1 stop bit, no parity
Maximum Depth	**2000m**

CAUTION - The maximum deployment depth will be limited by the measurement range of the optional pressure sensor, if installed.

Sea-Bird Electronics, Inc.
1808 136th Place NE, Bellevue, Washington 98005 USA
Website: http://www.seabird.com

Phone: (425) 643-9866
FAX: (425) 643-9954
Email: seabird@seabird.com

SBE Pressure Test Certificate

Test Date: **6/13/2005** Description **SBE-37 Microcat**

Job Number: **39208** Customer Name **ANTARCTIC CRC/U TASMANIA**

SBE Sensor Information:

Model Number: **37**
Serial Number: **3883**

Pressure Sensor Information:

Sensor Type: **None**
Sensor Serial Number: **None**
Sensor Rating: **0**

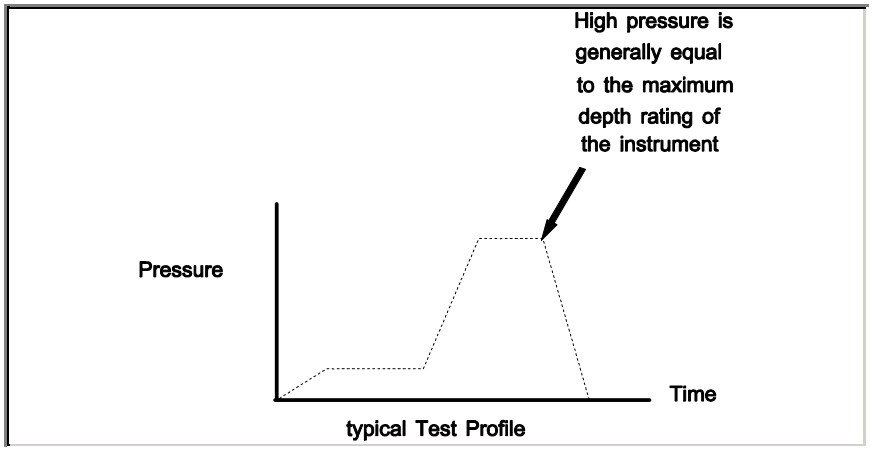
Pressure Test Protocol:

Low Pressure Test: **50** PSI Held For **15** Minutes

High Pressure Test: **2900** PSI Held For **15** Minutes

Passed Test: ☑

Tested By: PCC



typical Test Profile

SEA-BIRD ELECTRONICS, INC.

1808 136th Place N.E., Bellevue, Washington, 98005 USA

Phone: (425) 643 - 9866 Fax (425) 643 - 9954 Email: seabird@seabird.com

SENSOR SERIAL NUMBER: 3883
CALIBRATION DATE: 24-Jun-05

SBE 37 TEMPERATURE CALIBRATION DATA
ITS-90 TEMPERATURE SCALE

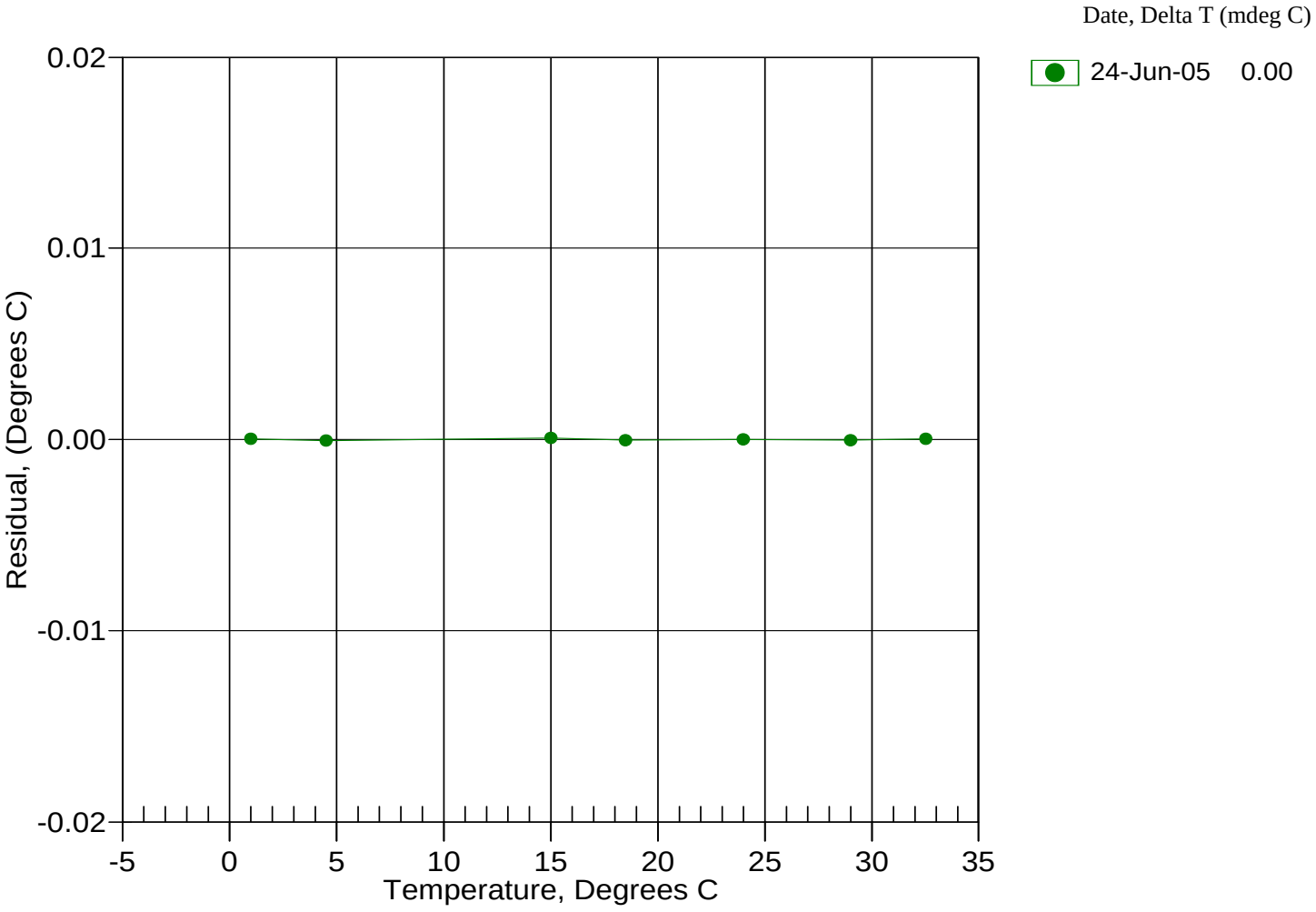
ITS-90 COEFFICIENTS

a0 = -3.190061e-005
a1 = 2.815857e-004
a2 = -2.692385e-006
a3 = 1.656168e-007

BATH TEMP (ITS-90)	INSTRUMENT OUTPUT	INST TEMP (ITS-90)	RESIDUAL (ITS-90)
1.0000	640923.2	1.0000	0.0000
4.5000	549227.6	4.4999	-0.0001
15.0000	352269.4	15.0001	0.0001
18.4999	305646.9	18.4999	-0.0000
24.0000	245950.7	24.0000	-0.0000
29.0000	203064.3	29.0000	-0.0000
32.5000	178151.1	32.5000	0.0000

Temperature ITS-90 = $1/\{a0 + a1[ln(n)] + a2[ln^2(n)] + a3[ln^3(n)]\} - 273.15$ (°C)

Residual = instrument temperature - bath temperature



SEA-BIRD ELECTRONICS, INC.

1808 136th Place N.E., Bellevue, Washington, 98005 USA

Phone: (425) 643 - 9866 Fax (425) 643 - 9954 Email: seabird@seabird.com

SENSOR SERIAL NUMBER: 3883
CALIBRATION DATE: 24-Jun-05

SBE 37 CONDUCTIVITY CALIBRATION DATA
PSS 1978: C(35,15,0) = 4.2914 Siemens/meter

COEFFICIENTS:

```
g = -1.022959e+000
h =  1.355084e-001
i = -1.644028e-004
j =  3.216407e-005

CPcor = -9.5700e-008
CTcor =  3.2500e-006
WBOTC = -1.3875e-005
```

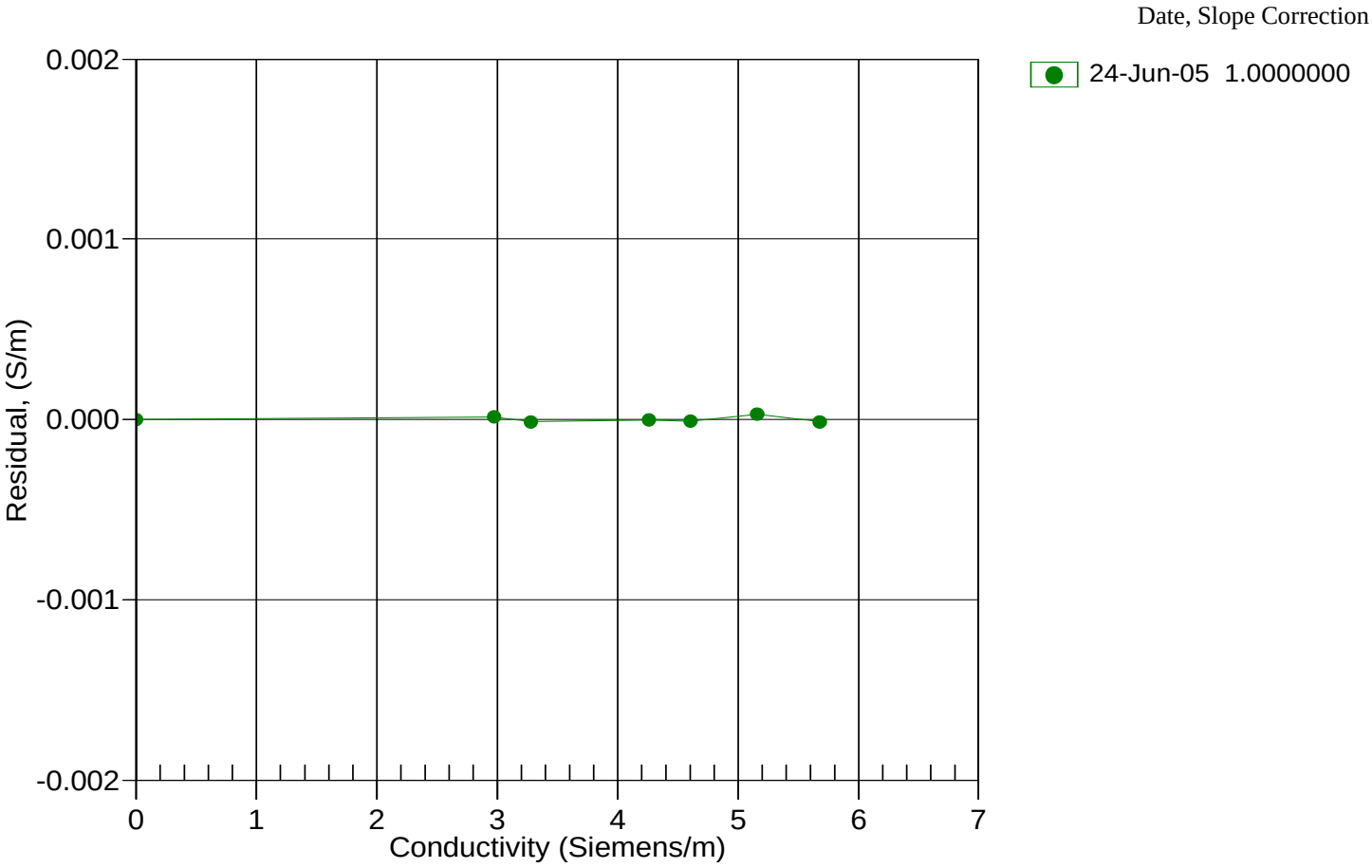
BATH TEMP (ITS-90)	BATH SAL (PSU)	BATH COND (Siemens/m)	INST FREQ (Hz)	INST COND (Siemens/m)	RESIDUAL (Siemens/m)
22.0000	0.0000	0.00000	2750.09	0.00000	0.00000
1.0000	34.7492	2.97078	5427.78	2.97079	0.00001
4.5000	34.7294	3.27733	5631.58	3.27732	-0.00001
15.0000	34.6852	4.25723	6237.83	4.25723	-0.00000
18.4999	34.6758	4.60173	6437.14	4.60172	-0.00001
24.0000	34.6645	5.15854	6746.58	5.15856	0.00003
29.0000	34.6586	5.67939	7023.35	5.67938	-0.00001

f = INST FREQ * sqrt(1.0 + WBOTC * t) / 1000.0

Conductivity = $(g + hf^2 + if^3 + jf^4) / (1 + \delta t + \varepsilon p)$ Siemens/meter

t = temperature[°C)]; p = pressure[decibars]; δ = CTcor; ε = CPcor;

Residual = instrument conductivity - bath conductivity



SEA-BIRD ELECTRONICS, INC.

1808 136th Place N.E., Bellevue, Washington, 98005 USA

Phone: (425) 643 - 9866 Fax (425) 643 - 9954 Email: seabird@seabird.com

SENSOR SERIAL NUMBER: 3883
CALIBRATION DATE: 24-Jun-05

SBE 37 RTC CALIBRATION DATA
ITS-90 TEMPERATURE SCALE

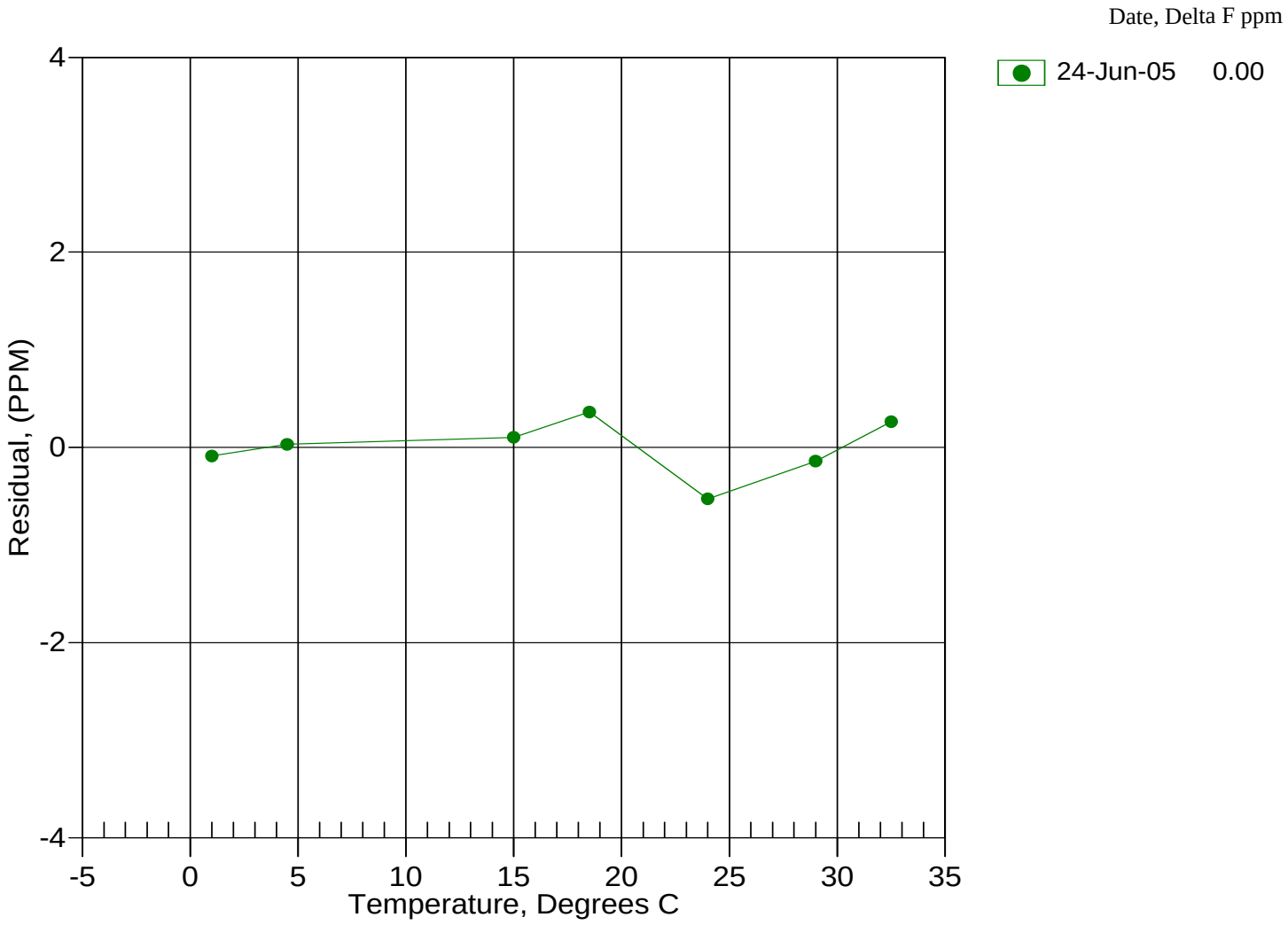
COEFFICIENTS:

```
rtca0 =  9.999803e-001
rtca1 =  1.623131e-006
rtca2 = -2.916839e-008
```

BATH TEMP (ITS-90)	RTC FREQ (Hz)	COMPUTED FREQ (Hz)	RESIDUAL (PPM)
1.0000	0.9999820	0.9999819	-0.1
4.5000	0.9999870	0.9999870	0.0
15.0000	0.9999980	0.9999981	0.1
18.4999	1.0000000	1.0000004	0.4
24.0000	1.0000030	1.0000025	-0.5
29.0000	1.0000030	1.0000029	-0.1
32.5000	1.0000020	1.0000023	0.3

RTC frequency = rtca0 + rtca1 * t + rtca2 * t^2

Residual = (Computed RTC frequency - Measured RTC frequency) * 1e6



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1808 136th Place N.E., Bellevue, Washington, 98005 USA
Phone: (425) 643 - 9866 Fax (425) 643 - 9954 Email: seabird@seabird.com

SENSOR SERIAL NUMBER: 3883
CALIBRATION DATE: 14-Oct-04

SBE 37 PRESSURE CALIBRATION DATA
3000 psia S/N 6998

COEFFICIENTS:

```
PA0 =    5.041017e-001
PA1 =    1.379747e-001
PA2 =   -3.695607e-008

PTCA0 = -8.010872e+001
PTCA1 =  2.437599e-001
PTCA2 = -3.792095e-003
PTCB0 =  2.499488e+001
PTCB1 = -2.500000e-005
PTCB2 =  0.000000e+000
```

PRESSURE SPAN CALIBRATION

PRESSURE PSIA	INST OUTPUT	TEMP ITS90	COMPUTED PRESSURE	ERROR %FS
14.73	27.1	23.8	14.79	0.00
614.91	4381.7	23.8	614.90	-0.00
1215.05	8746.2	23.8	1214.96	-0.00
1815.11	13121.6	23.8	1815.11	-0.00
2415.05	17506.2	23.8	2415.10	0.00
3015.14	21901.0	23.8	3015.06	-0.00
2415.08	17506.8	23.8	2415.18	0.00
1815.06	13121.5	23.8	1815.09	0.00
1215.05	8746.6	23.8	1215.01	-0.00
614.84	4381.3	23.8	614.84	0.00
14.73	26.5	23.8	14.71	-0.00

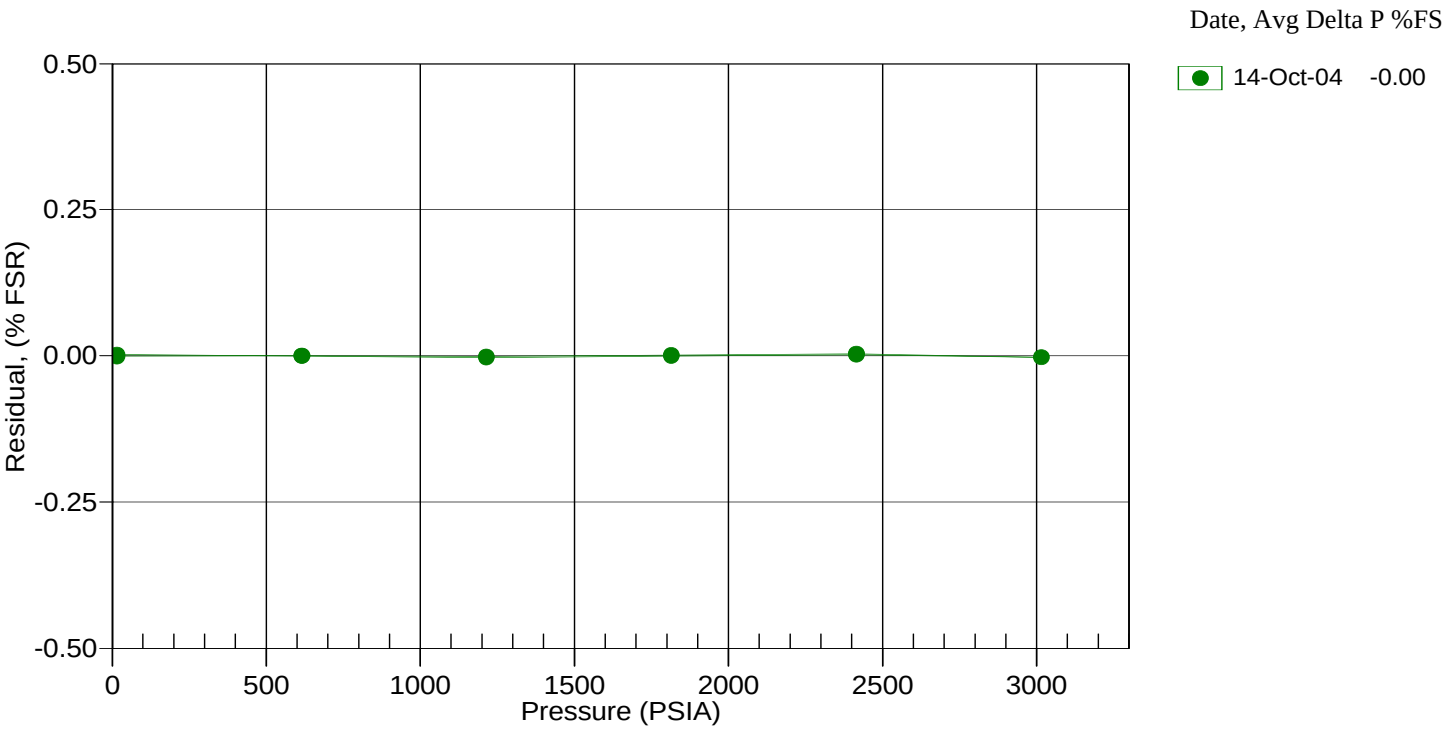
THERMAL CORRECTION

TEMP ITS90	INST OUTPUT	TEMP ITS90	SPAN MV
32.50	32.40	-5.00	25.00
29.00	32.60	35.00	24.99
24.00	32.40		
18.50	31.80		
15.00	31.30		
4.50	29.60		
1.00	28.90		

x = pressure output - PTCA0 - PTCA1 * t - PTCA2 * t^2

$n = x * PTCB0 / (PTCB0 + PTCB1 * t + PTCB2 * t^2)$

pressure (psia) = PA0 + PA1 * n + PA2 * n^2



SBE37-IM MicroCAT

Conductivity & Temperature Recorder
with Inductive Modem

Instrument Configuration:

Serial Number	37IM39208-4054
Pressure Sensor	1000 dBar Druck, SN 7433
Firmware Version	2.3a
Memory	2048Kb
Conductivity Range	0-7 S/m
Microcat ID#	02
DPSK Baud Rate	1200 (Microcat <-> SIM)
SIM Baud Rate	9600, 8 data bits, 1 stop bit, no parity
Maximum Depth	**1000m**

CAUTION - The maximum deployment depth will be limited by the measurement range of the optional pressure sensor, if installed.

Sea-Bird Electronics, Inc.
1808 136th Place NE, Bellevue, Washington 98005 USA
Website: http://www.seabird.com

Phone: (425) 643-9866
FAX: (425) 643-9954
Email: seabird@seabird.com

SBE Pressure Test Certificate

Test Date: **6/13/2005** Description **SBE-37 Microcat**

Job Number: **39208** Customer Name **ANTARCTIC CRC/U TASMANIA**

SBE Sensor Information:

Model Number: **37**
Serial Number: **4054**

Pressure Sensor Information:

Sensor Type: **None**
Sensor Serial Number: **None**
Sensor Rating: **0**

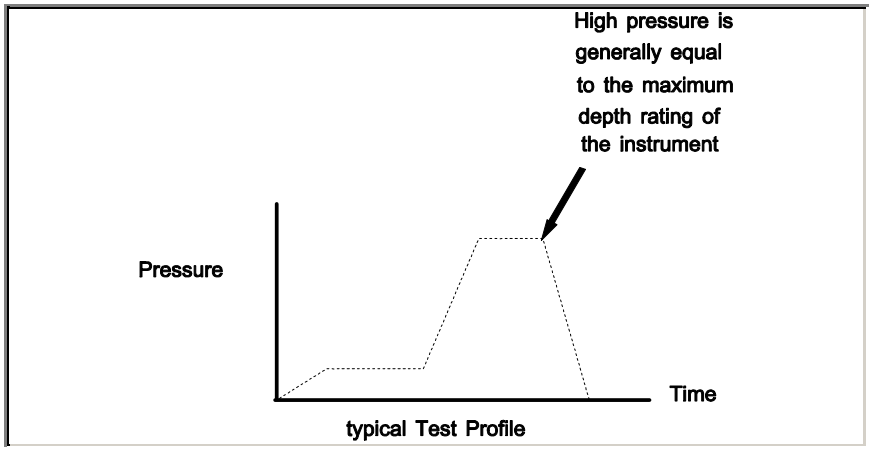
Pressure Test Protocol:

Low Pressure Test: **50** PSI Held For **15** Minutes

High Pressure Test: **1450** PSI Held For **15** Minutes

Passed Test: ☑

Tested By: PCC



typical Test Profile

SEA-BIRD ELECTRONICS, INC.

1808 136th Place N.E., Bellevue, Washington, 98005 USA

Phone: (425) 643 - 9866 Fax (425) 643 - 9954 Email: seabird@seabird.com

SENSOR SERIAL NUMBER: 4054
CALIBRATION DATE: 24-Jun-05

SBE 37 TEMPERATURE CALIBRATION DATA
ITS-90 TEMPERATURE SCALE

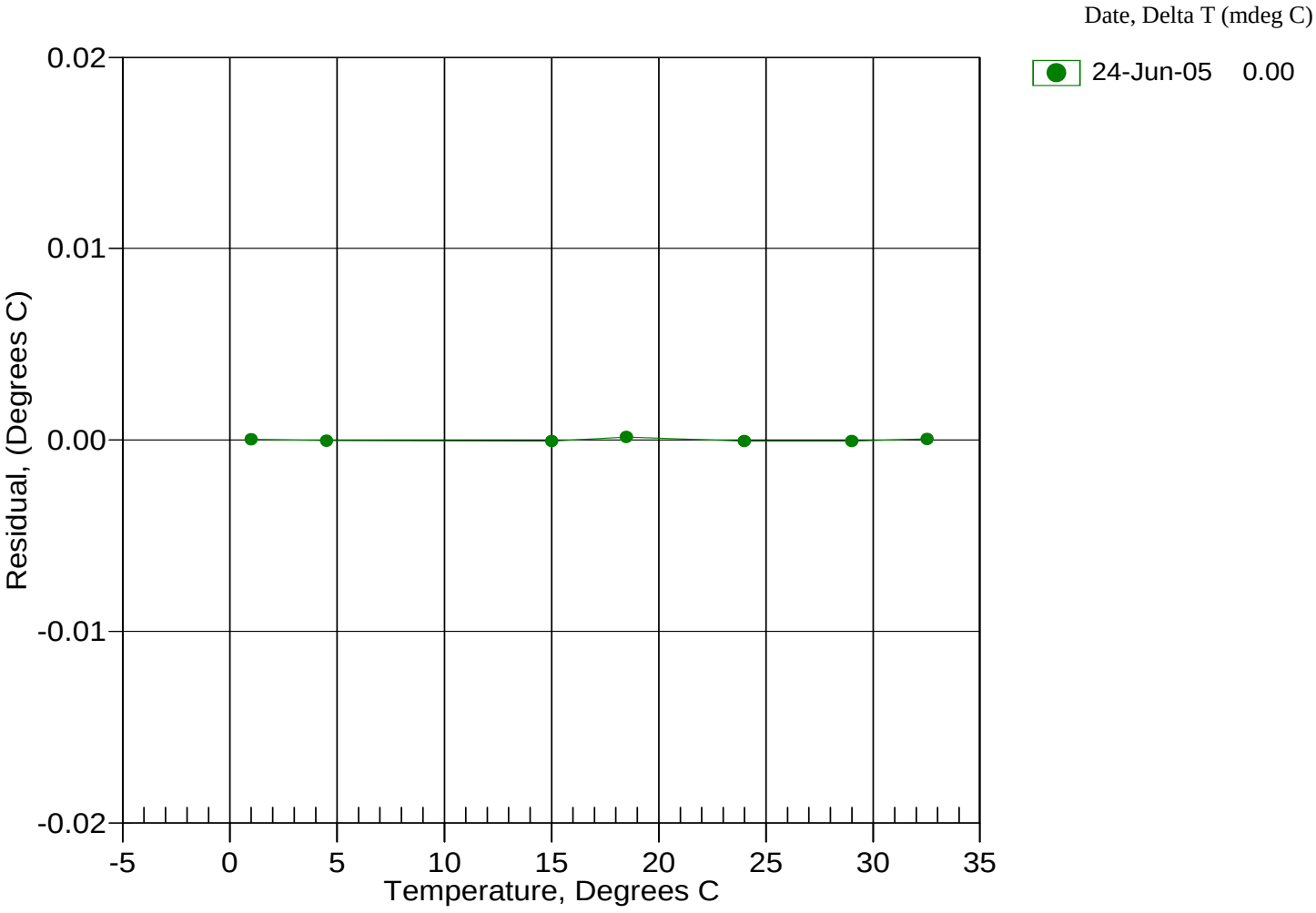
ITS-90 COEFFICIENTS

a0 = -5.694242e-005
a1 = 2.858881e-004
a2 = -3.060785e-006
a3 = 1.729209e-007

BATH TEMP (ITS-90)	INSTRUMENT OUTPUT	INST TEMP (ITS-90)	RESIDUAL (ITS-90)
1.0000	676213.1	1.0000	0.0000
4.5000	579103.0	4.5000	-0.0000
15.0000	370769.1	14.9999	-0.0001
18.4999	321512.7	18.5001	0.0002
24.0000	258499.6	23.9999	-0.0001
29.0000	213267.0	28.9999	-0.0001
32.5000	187008.4	32.5000	0.0000

Temperature ITS-90 = $1/\{a0 + a1[ln(n)] + a2[ln^2(n)] + a3[ln^3(n)]\} - 273.15$ (°C)

Residual = instrument temperature - bath temperature



SEA-BIRD ELECTRONICS, INC.

1808 136th Place N.E., Bellevue, Washington, 98005 USA

Phone: (425) 643 - 9866 Fax (425) 643 - 9954 Email: seabird@seabird.com

SENSOR SERIAL NUMBER: 4054
CALIBRATION DATE: 24-Jun-05

SBE 37 CONDUCTIVITY CALIBRATION DATA
PSS 1978: C(35,15,0) = 4.2914 Siemens/meter

COEFFICIENTS:

```
g = -1.026244e+000
h =  1.506195e-001
i = -1.721474e-004
j =  3.855986e-005

CPcor = -9.5700e-008
CTcor =  3.2500e-006
WBOTC = -1.4383e-005
```

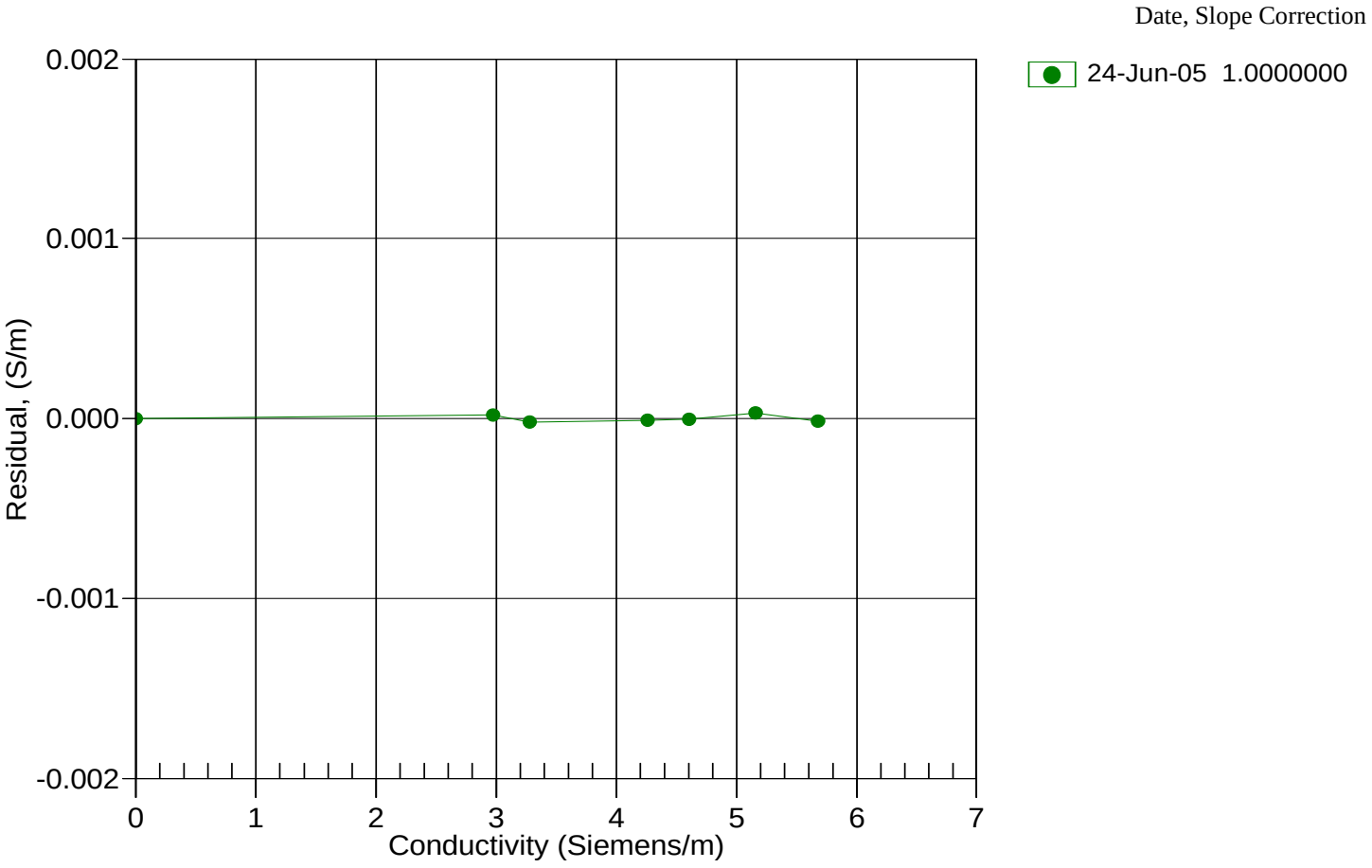
BATH TEMP (ITS-90)	BATH SAL (PSU)	BATH COND (Siemens/m)	INST FREQ (Hz)	INST COND (Siemens/m)	RESIDUAL (Siemens/m)
22.0000	0.0000	0.00000	2612.30	0.00000	0.00000
1.0000	34.7492	2.97078	5149.16	2.97080	0.00002
4.5000	34.7294	3.27733	5342.31	3.27731	-0.00002
15.0000	34.6852	4.25723	5916.94	4.25722	-0.00001
18.4999	34.6758	4.60173	6105.87	4.60172	-0.00000
24.0000	34.6645	5.15854	6399.19	5.15857	0.00003
29.0000	34.6586	5.67939	6661.55	5.67937	-0.00002

f = INST FREQ * sqrt(1.0 + WBOTC * t) / 1000.0

Conductivity = $(g + hf^2 + if^3 + jf^4) / (1 + \delta t + \varepsilon p)$ Siemens/meter

t = temperature[°C)]; p = pressure[decibars]; δ = CTcor; ε = CPcor;

Residual = instrument conductivity - bath conductivity



SEA-BIRD ELECTRONICS, INC.

1808 136th Place N.E., Bellevue, Washington, 98005 USA

Phone: (425) 643 - 9866 Fax (425) 643 - 9954 Email: seabird@seabird.com

SENSOR SERIAL NUMBER: 4054
CALIBRATION DATE: 24-Jun-05

SBE 37 RTC CALIBRATION DATA
ITS-90 TEMPERATURE SCALE

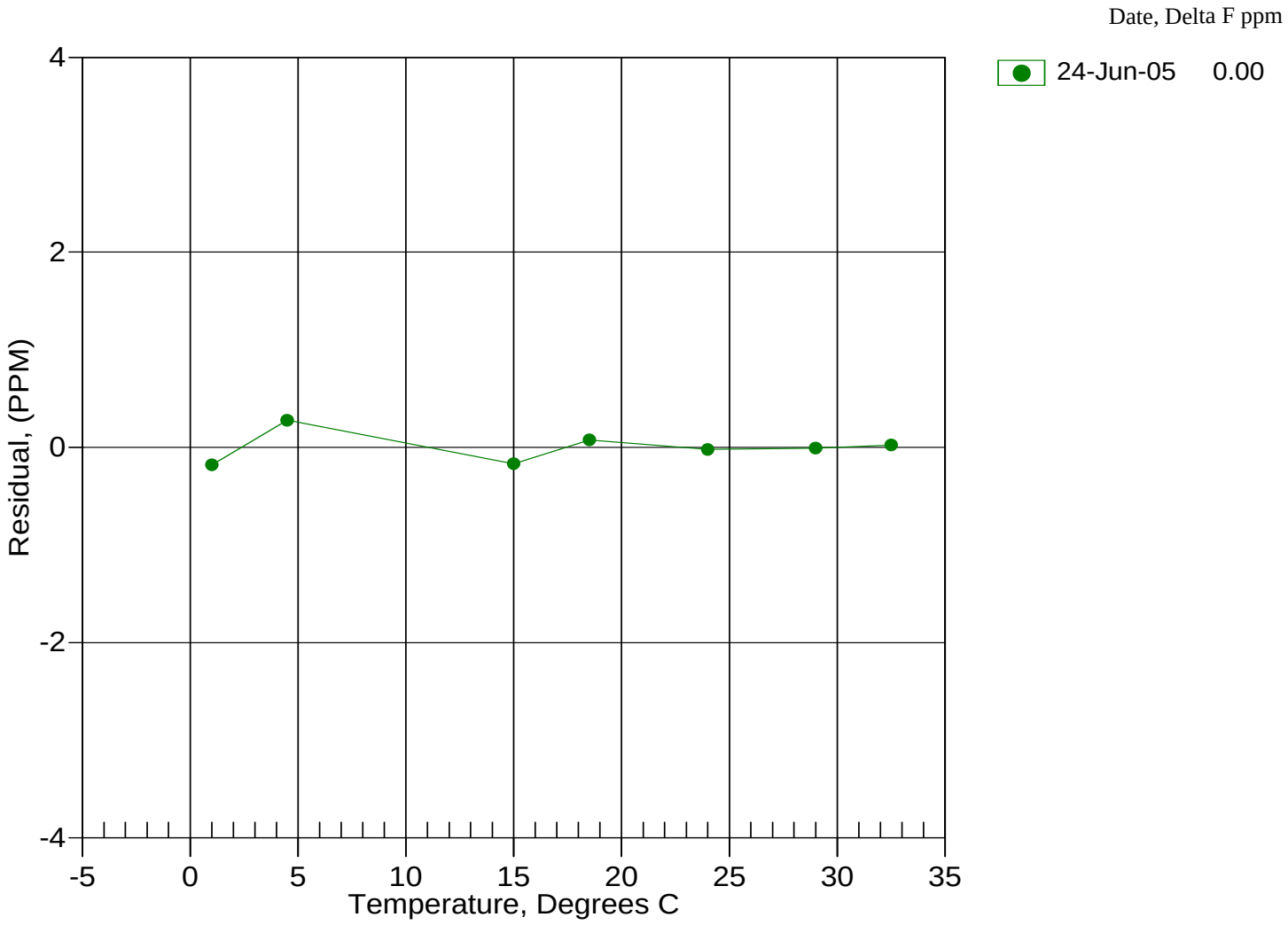
COEFFICIENTS:

```
rtca0 =  9.999871e-001
rtca1 =  1.739719e-006
rtca2 = -3.278587e-008
```

BATH TEMP (ITS-90)	RTC FREQ (Hz)	COMPUTED FREQ (Hz)	RESIDUAL (PPM)
1.0000	0.9999890	0.9999888	-0.2
4.5000	0.9999940	0.9999943	0.3
15.0000	1.0000060	1.0000058	-0.2
18.4999	1.0000080	1.0000081	0.1
24.0000	1.0000100	1.0000100	-0.0
29.0000	1.0000100	1.0000100	-0.0
32.5000	1.0000090	1.0000090	0.0

RTC frequency = rtca0 + rtca1 * t + rtca2 * t^2

Residual = (Computed RTC frequency - Measured RTC frequency) * 1e6



SEA-BIRD ELECTRONICS, INC.

1808 136th Place N.E., Bellevue, Washington, 98005 USA

Phone: (425) 643 - 9866 Fax (425) 643 - 9954 Email: seabird@seabird.com

SENSOR SERIAL NUMBER: 4054
CALIBRATION DATE: 22-Jun-05

SBE 37 PRESSURE CALIBRATION DATA
1450 psia S/N 7433

COEFFICIENTS:

```
PA0 =   -1.926511e-002
PA1 =    6.833797e-002
PA2 =   -2.760009e-009

PTCA0 = -1.939810e+002
PTCA1 = -1.440972e-001
PTCA2 =  5.918220e-003
PTCB0 =  2.496775e+001
PTCB1 = -1.050000e-003
PTCB2 =  0.000000e+000
```

PRESSURE SPAN CALIBRATION

PRESSURE PSIA	INST OUTPUT	TEMP ITS90	COMPUTED PRESSURE	ERROR %FS
14.68	21.0	22.2	14.71	0.00
314.89	4408.6	22.2	314.77	-0.01
614.85	8796.7	22.1	614.75	-0.01
914.80	13187.2	22.1	914.80	0.00
1214.93	17578.7	22.1	1214.81	-0.01
1514.84	21972.2	22.0	1514.84	-0.00
1214.78	17579.1	21.8	1214.82	0.00
914.65	13187.7	21.8	914.82	0.01
614.74	8797.4	21.9	614.80	0.00
314.81	4409.1	22.0	314.80	-0.00
14.68	21.3	22.0	14.73	0.00

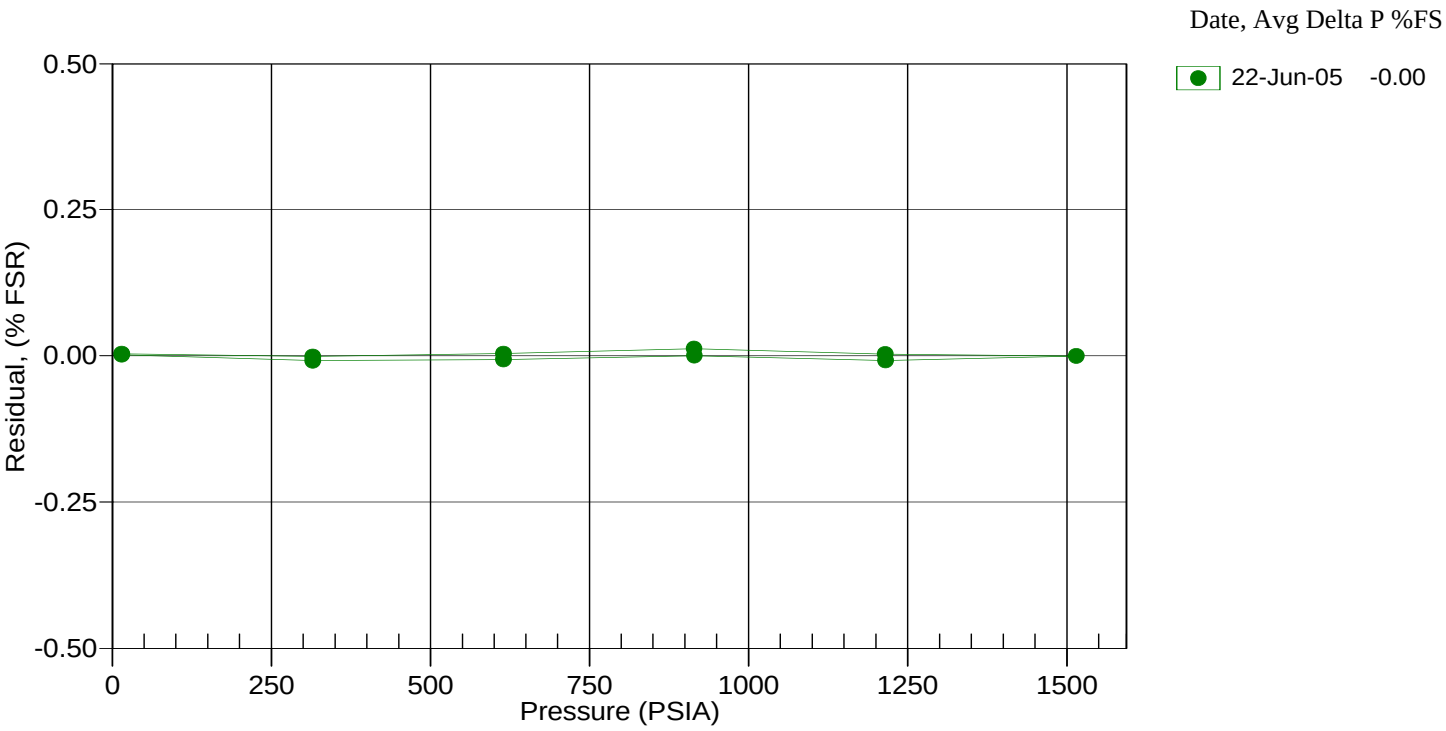
THERMAL CORRECTION

TEMP ITS90	INST OUTPUT	TEMP ITS90	SPAN MV
32.50	33.20	-5.00	24.97
29.00	32.40	35.00	24.93
24.00	31.60		
18.50	30.90		
15.00	30.80		
4.50	31.20		
1.00	31.40		

x = pressure output - PTCA0 - PTCA1 * t - PTCA2 * t^2

n = x * PTCB0 / (PTCB0 + PTCB1 * t + PTCB2 * t^2)

pressure (psia) = PA0 + PA1 * n + PA2 * n^2



SBE37-IM MicroCAT

Conductivity & Temperature Recorder
with Inductive Modem

Instrument Configuration:

Serial Number	37IM39208-4055
Pressure Sensor	1000 dBar Druck, SN 7434
Firmware Version	2.3a
Memory	2048Kb
Conductivity Range	0-7 S/m
Microcat ID#	03
DPSK Baud Rate	1200 (Microcat <-> SIM)
SIM Baud Rate	9600, 8 data bits, 1 stop bit, no parity
Maximum Depth	**1000m**

CAUTION - The maximum deployment depth will be limited by the measurement range of the optional pressure sensor, if installed.

Sea-Bird Electronics, Inc.
1808 136th Place NE, Bellevue, Washington 98005 USA
Website: http://www.seabird.com

Phone: (425) 643-9866
FAX: (425) 643-9954
Email: seabird@seabird.com

SBE Pressure Test Certificate

Test Date: **6/13/2005** Description **SBE-37 Microcat**

Job Number: **39208** Customer Name **ANTARCTIC CRC/U TASMANIA**

SBE Sensor Information:

Model Number: **37**
Serial Number: **4055**

Pressure Sensor Information:

Sensor Type: **None**
Sensor Serial Number: **None**
Sensor Rating: **0**

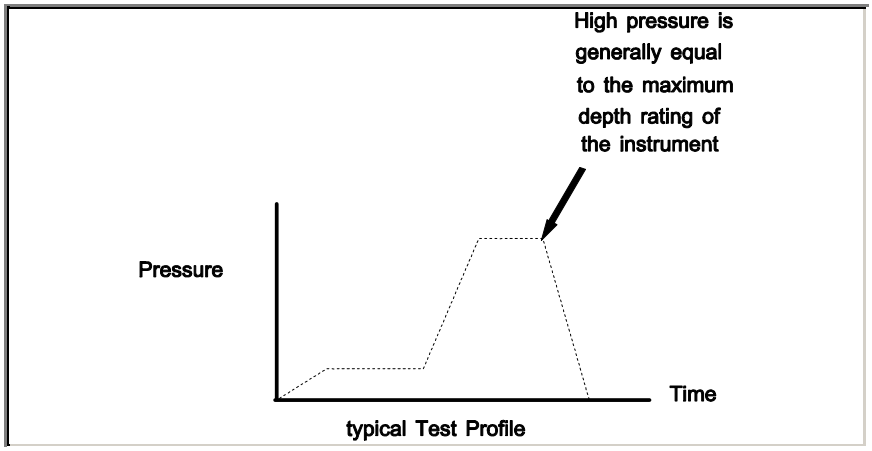
Pressure Test Protocol:

Low Pressure Test: **50** PSI Held For **15** Minutes

High Pressure Test: **1450** PSI Held For **15** Minutes

Passed Test: ☑

Tested By: PCC



SEA-BIRD ELECTRONICS, INC.

1808 136th Place N.E., Bellevue, Washington, 98005 USA

Phone: (425) 643 - 9866 Fax (425) 643 - 9954 Email: seabird@seabird.com

SENSOR SERIAL NUMBER: 4055
CALIBRATION DATE: 24-Jun-05

SBE 37 TEMPERATURE CALIBRATION DATA
ITS-90 TEMPERATURE SCALE

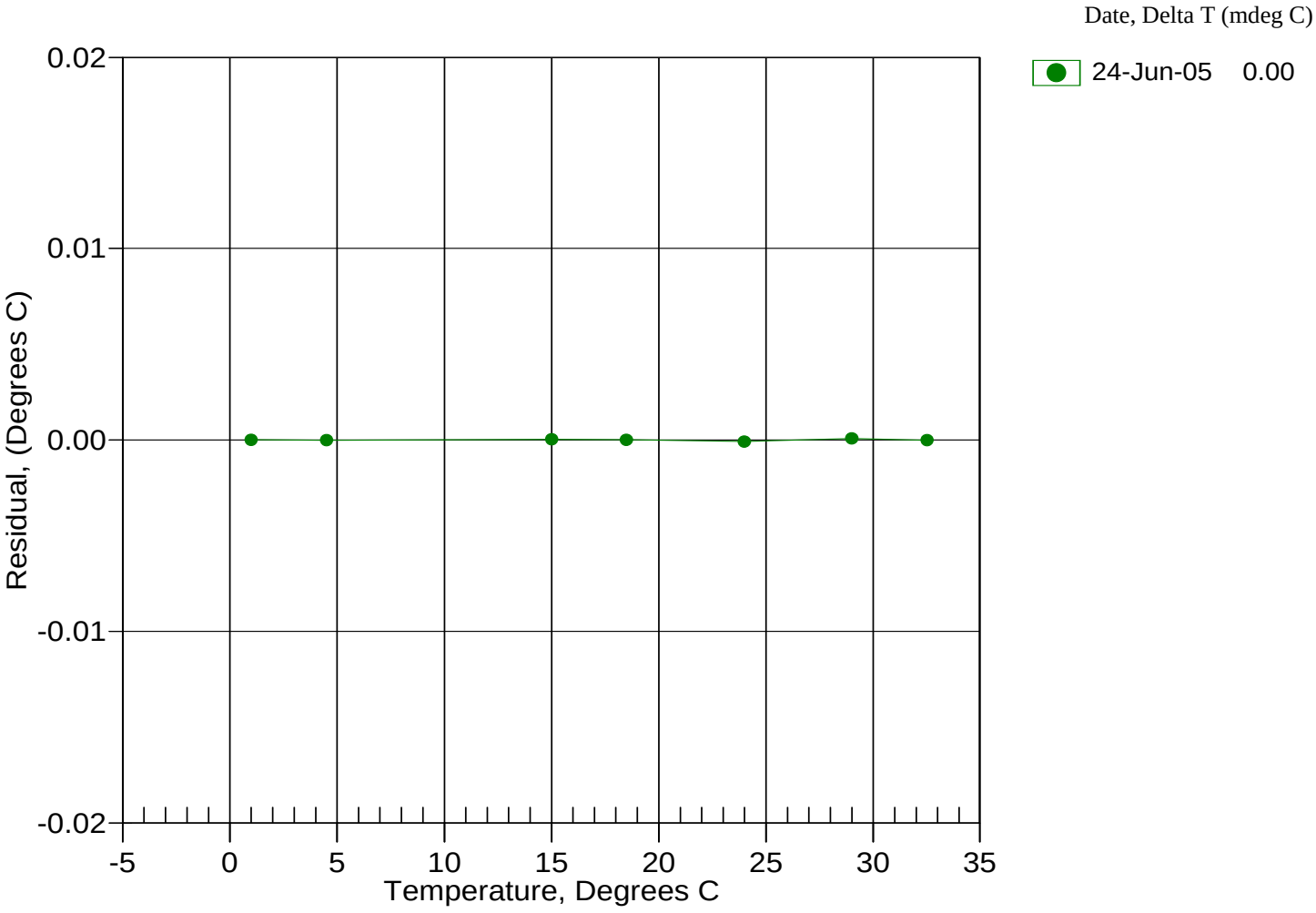
ITS-90 COEFFICIENTS

a0 = -7.127527e-005
a1 = 2.904089e-004
a2 = -3.417971e-006
a3 = 1.827576e-007

BATH TEMP (ITS-90)	INSTRUMENT OUTPUT	INST TEMP (ITS-90)	RESIDUAL (ITS-90)
1.0000	663189.5	1.0000	0.0000
4.5000	567976.6	4.5000	-0.0000
15.0000	363688.1	15.0000	0.0000
18.4999	315386.5	18.4999	0.0000
24.0000	253586.3	23.9999	-0.0001
29.0000	209222.1	29.0001	0.0001
32.5000	183469.0	32.5000	-0.0000

Temperature ITS-90 = $1/\{a0 + a1[ln(n)] + a2[ln^2(n)] + a3[ln^3(n)]\} - 273.15$ (°C)

Residual = instrument temperature - bath temperature



SEA-BIRD ELECTRONICS, INC.

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Phone: (425) 643 - 9866 Fax (425) 643 - 9954 Email: seabird@seabird.com

SENSOR SERIAL NUMBER: 4055
CALIBRATION DATE: 24-Jun-05

SBE 37 CONDUCTIVITY CALIBRATION DATA
PSS 1978: C(35,15,0) = 4.2914 Siemens/meter

COEFFICIENTS:

```
g = -1.026250e+000
h =  1.443525e-001
i = -2.216091e-004
j =  3.992332e-005

CPcor = -9.5700e-008
CTcor =  3.2500e-006
WBOTC =  1.6864e-005
```

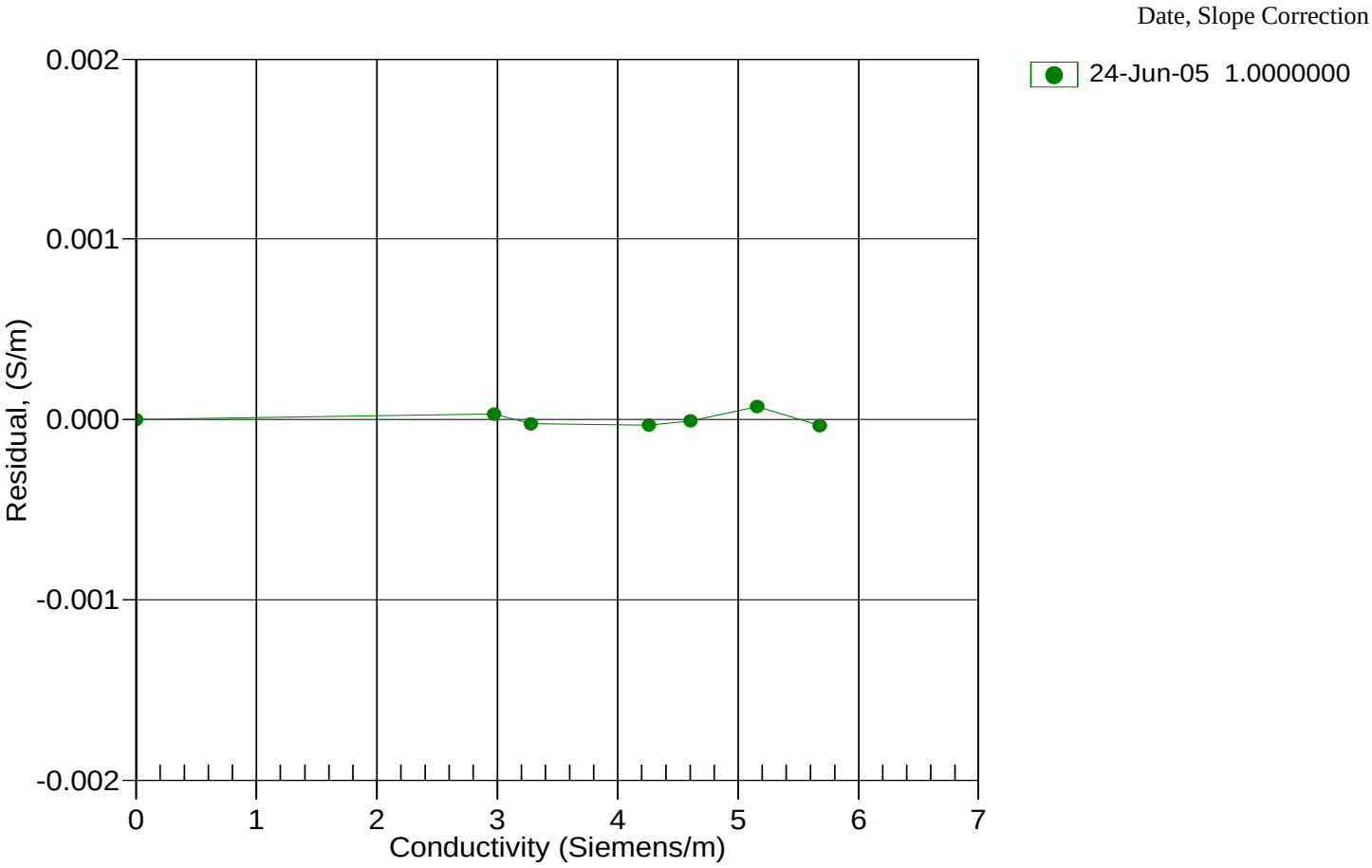
BATH TEMP (ITS-90)	BATH SAL (PSU)	BATH COND (Siemens/m)	INST FREQ (Hz)	INST COND (Siemens/m)	RESIDUAL (Siemens/m)
22.0000	0.0000	0.00000	2668.68	0.00000	0.00000
1.0000	34.7492	2.97078	5263.15	2.97081	0.00003
4.5000	34.7294	3.27733	5460.31	3.27731	-0.00002
15.0000	34.6852	4.25723	6046.72	4.25720	-0.00003
18.4999	34.6758	4.60173	6239.48	4.60172	-0.00001
24.0000	34.6645	5.15854	6538.69	5.15861	0.00007
29.0000	34.6586	5.67939	6806.20	5.67936	-0.00003

f = INST FREQ * sqrt(1.0 + WBOTC * t) / 1000.0

Conductivity = $(g + hf^2 + if^3 + jf^4) / (1 + \delta t + \varepsilon p)$ Siemens/meter

t = temperature[°C)]; p = pressure[decibars]; δ = CTcor; ε = CPcor;

Residual = instrument conductivity - bath conductivity



SEA-BIRD ELECTRONICS, INC.

1808 136th Place N.E., Bellevue, Washington, 98005 USA

Phone: (425) 643 - 9866 Fax (425) 643 - 9954 Email: seabird@seabird.com

SENSOR SERIAL NUMBER: 4055
CALIBRATION DATE: 24-Jun-05

SBE 37 RTC CALIBRATION DATA
ITS-90 TEMPERATURE SCALE

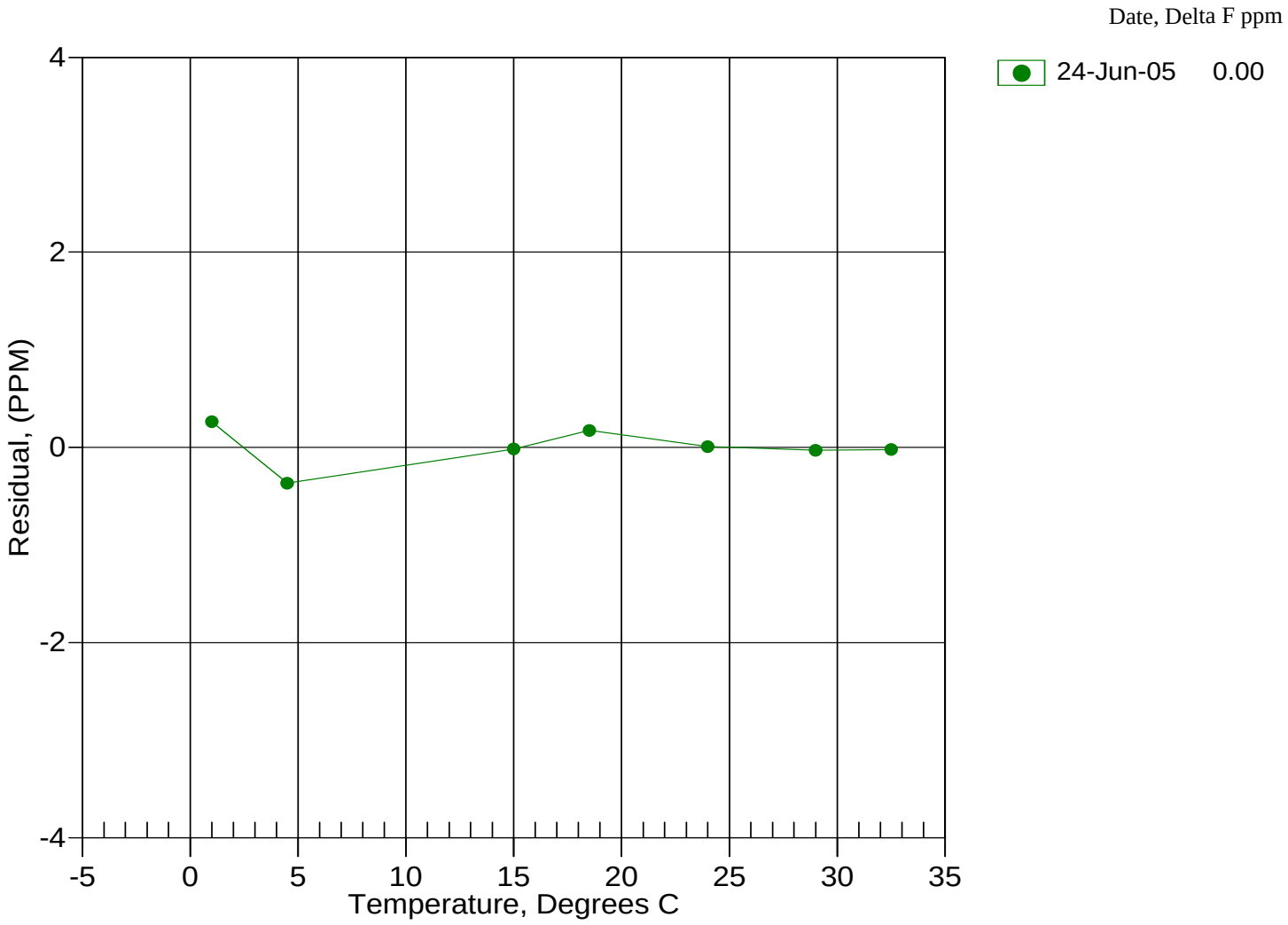
COEFFICIENTS:

```
rtca0 =  9.999736e-001
rtca1 =  1.714086e-006
rtca2 = -3.248331e-008
```

BATH TEMP (ITS-90)	RTC FREQ (Hz)	COMPUTED FREQ (Hz)	RESIDUAL (PPM)
1.0000	0.9999750	0.9999753	0.3
4.5000	0.9999810	0.9999806	-0.4
15.0000	0.9999920	0.9999920	-0.0
18.4999	0.9999940	0.9999942	0.2
24.0000	0.9999960	0.9999960	0.0
29.0000	0.9999960	0.9999960	-0.0
32.5000	0.9999950	0.9999950	-0.0

RTC frequency = rtca0 + rtca1 * t + rtca2 * t^2

Residual = (Computed RTC frequency - Measured RTC frequency) * 1e6



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SENSOR SERIAL NUMBER: 4055
CALIBRATION DATE: 14-Jun-05

SBE 37 PRESSURE CALIBRATION DATA
1450 psia S/N 7434

COEFFICIENTS:

```
PA0 =   -1.579150e+000
PA1 =    6.842592e-002
PA2 =   -2.899693e-009

PTCA0 = -3.653192e+002
PTCA1 = -1.507080e+000
PTCA2 =  2.135777e-002
PTCB0 =  2.472238e+001
PTCB1 = -1.725000e-003
PTCB2 =  0.000000e+000
```

PRESSURE SPAN CALIBRATION

PRESSURE PSIA	INST OUTPUT	TEMP ITS90	COMPUTED PRESSURE	ERROR %FS
14.60	-151.7	22.4	14.64	0.00
314.80	4227.6	22.4	314.70	-0.01
614.73	8606.6	22.4	614.64	-0.01
914.68	12987.9	22.4	914.62	-0.00
1214.72	17370.8	22.5	1214.60	-0.01
1514.70	21757.2	22.6	1514.73	0.00
1214.65	17371.9	22.6	1214.69	0.00
914.55	12988.6	22.6	914.68	0.01
614.60	8607.3	22.7	614.71	0.01
314.67	4227.6	22.5	314.71	0.00
14.60	-152.5	22.4	14.58	-0.00

THERMAL CORRECTION

TEMP ITS90	INST OUTPUT	TEMP ITS90	SPAN MV
32.50	-143.10	-5.00	24.73
29.00	-142.20	35.00	24.66
24.00	-140.30		
18.50	-137.20		
15.00	-134.50		
4.50	-122.80		
1.00	-118.10		

x = pressure output - PTCA0 - PTCA1 * t - PTCA2 * t^2

$n = x * PTCB0 / (PTCB0 + PTCB1 * t + PTCB2 * t^2)$

pressure (psia) = PA0 + PA1 * n + PA2 * n^2

